



EC 102: Fundamentals of Electronics

Course Code	EC 102
Course Name	Fundamentals of Electronics
L-T-P-C	3-1-2-4
Prerequisites	NIL Sem. Both
Approval	5th Senate (Oct 2013)

Course Contents

- **Review of conductors, semiconductors, and insulators; Drift and diffusion currents;** p-n junction; junction under forward and reverse bias; circuit models; diode applications: rectifier, clipper, clamper; Zener diode regulator; simple power supply with capacitor filter and zener regulator.
- **Bipolar Junction Transistor:** structure and operation, various configurations, input and output characteristics, BJT as amplifier, DC analysis of various biasing circuits, biasing stability.
- **Field Effect Transistors:** JFET, depletion-mode and enhancement-mode MOSFETs, FET biasing, FET as an amplifier.
- **Small- signal analysis of BJTs and FETs:** h-parameter model of BJT, small-signal analysis of BJT amplifier circuits, frequency response of RC-coupled BJT and FET amplifiers.
- **Amplifiers:** cascade connection, current mirror, differential amplifier, operational amplifier, op-amp applications, power amplifiers, feedback in amplifiers.
- **Oscillators:** Barkhausen criterion, damped oscillations in LC circuits, audio and rf oscillators.
- **Digital Electronics:** Combinational Circuits – adder, de-coder, encoder, multiplexer, demultiplexer; Sequential Circuits – flip-flops, counters, and shift registers; ADC and DAC.

References

1. Boylestad R. L. and Nashelsky L., **Electronic Devices and Circuit Theory**, 9th Edition, Pearson Education.
2. Millman J. and Halkias C. C., **Electronic Devices and Circuit**, McGraw-Hill.
3. Millman J. and Halkias C. C., **Integrated Electronics**, Tata McGraw- Hill.
4. Nagrath I. J., **Electronics – Analog and Digital**, Prentice Hall of India.
5. Santiram K., **Basic Electronics**, Prentice Hall of India.